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Timeflow Gravity: a thermodynamic theory of spacetime

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Abstract

This paper presents Timeflow Gravity, a framework in which gravity emerges as a thermodynamic, entropic force driven by the wave mechanics of a continuous $U(1)$ spacetime medium. Building on T. Jacobson's derivation of Einstein's equations from the Clausius relation, we derive the Einstein field equations as an emergent macroscopic equation of state. Within this framework, we interpret dark matter and dark energy effects as wave-mechanical projections of the underlying phase space. At galactic scales, constructive phase interference between baryonic matter and the vacuum recovers MOND-like dynamics. By incorporating kinematic phase decoherence, the theory offers a potential resolution to MOND's mass discrepancy in galaxy clusters, as well as the spatial offset observed in the Bullet Cluster. On cosmological scales, conservation of the one-dimensional topological phase boundary yields parameter-free matter and vacuum density parameters, offering a possible resolution to the cosmological coincidence problem. This result yields a dynamical dark energy equation of state ($w \approx -0.84$) consistent with recent Dark Energy Spectroscopic Instrument observations. Finally, we establish a falsifiability criterion: the intrinsic scatter of the Radial Acceleration Relation should systematically anti-correlate with the local macroscopic kinematic entropy of the system.

1. Introduction

General Relativity and its cosmological extension, the Λ CDM model, have achieved remarkable success in describing the large-scale structure and evolution of the Universe. However, this success relies on the introduction of two dominant but fundamentally unexplained components: dark matter and dark energy. Together, they are inferred to constitute approximately 95% of the total energy budget of the Universe, yet neither has been directly detected nor derived from established particle physics.

A series of persistent observational tensions suggests that these components may not represent fundamental physical substances, but could instead indicate a deeper incompleteness in our understanding of gravity. At galactic scales, the discrepancy between observed rotation curves and baryonic matter distributions is conventionally attributed to dark matter halos. However, the existence of tight empirical relations, such as the radial acceleration relation (RAR) [34], implies a strong and highly constrained coupling between baryonic matter and the effective gravitational field, which is not naturally expected in standard collisionless dark matter scenarios.

At the scale of galaxy clusters, observations such as the Bullet Cluster [15, 33] reveal a spatial separation between baryonic matter and gravitational lensing peaks. This phenomenon is widely interpreted as evidence for collisionless dark matter, yet it remains difficult to reconcile simultaneously with the success of modified gravity models at galactic scales, highlighting a persistent theoretical tension.

On cosmological scales, recent results from the Dark Energy Spectroscopic Instrument (DESI) [17, 22] provide evidence that the dark energy equation of state may deviate from a cosmological constant ($w = -1$), suggesting a dynamical component. This challenges one of the central assumptions of the Λ CDM model and sharpens the cosmological coincidence problem: why the densities of matter and dark energy are of the same order precisely in the present epoch [63].

Together, these tensions may suggest that dark matter and dark energy may not be missing substances but indications of a deeper principle—one rooted in the fundamental connection between space-time, energy, and gravity.

1.1. Time, gravity and thermodynamics

The path to the present framework is motivated by a fundamental tension in our current understanding of gravity: the absence of a well-defined, covariant notion of gravitational energy.

In General Relativity, energy-momentum conservation is expressed locally through the vanishing covariant divergence of the stress-energy tensor, $\nabla^\mu T_{\mu\nu} = 0$. However, this relation does not imply the existence of a globally conserved energy in a dynamical spacetime [14]. In particular, in the absence of a global timelike Killing vector, no invariant definition of total energy exists [58]. This leads to the well-known difficulty that gravitational energy cannot be localized in a coordinate-independent way.

Attempts to resolve this issue within the geometric framework of General Relativity have historically relied on the introduction of energy-momentum pseudotensors, beginning with Einstein [44] and later refined by Landau and Lifshitz [31]. However, these constructions are inherently coordinate-dependent and therefore lack direct physical observability. More recent approaches that attempt to introduce a genuine covariant gravitational stress-energy tensor, such as the Yilmaz theory of gravity [62], face significant theoretical and observational challenges, and have not led to a consistent alternative.

This persistent difficulty suggests that the problem may not lie in the mathematical formulation alone, but in the underlying physical interpretation of gravity. If gravitational dynamics cannot be consistently associated with a localizable energy density within a purely geometric framework, it is natural to consider whether gravity instead emerges from a deeper thermodynamic or informational structure.

The second insight arises from the intersection of gravity, quantum theory, and thermodynamics. In 1967, Andrei Sakharov [57] proposed the concept of induced gravity, in which spacetime curvature emerges as the macroscopic elasticity of the quantum vacuum, similar to fluid mechanics. Stephen Hawking's discovery of black hole radiation [26] and Jacob Bekenstein's formulation of black hole entropy [6] revealed a deep and unexpected thermodynamic structure underlying gravitational phenomena. In 1995, Theodore Jacobson demonstrated [29] that the Einstein field equations themselves can be derived from the Clausius relation, $\delta Q = TdS$, applied to local Rindler horizons, framing General Relativity as an equation of state. This result that the vacuum acts as a physical medium inspired formal mathematical frameworks like Einstein-ether theories [28], which explicitly introduce a dynamical preferred frame to capture this fluidic nature. Erik Verlinde [56] has advanced the view that gravity is an entropic force emerging from the statistical behavior of microscopic information, later extending this framework [55] to propose that phenomena attributed to dark matter are actually the emergent thermodynamic response of the background vacuum entropy. In parallel, Mordehai Milgrom's modified Newtonian dynamics (MOND) [35, 36] uncovered a characteristic acceleration scale, a_0 , which not only successfully describes many galactic rotation curves without invoking dark matter but also intriguingly matches cosmological parameters such as the Hubble constant ($a_0 \sim cH_0$) [37]. Furthermore, the deep connection between MOND's acceleration scale and the quantum vacuum has been explored via the Unruh effect [39], hinting at a fundamental thermodynamic origin for modified inertia or gravity.

Alongside MOND, a rich landscape of alternative gravity theories has emerged to address these astrophysical anomalies without invoking dark matter. Prominent examples include generalizations of the Einstein-Hilbert action such as $f(R)$ gravity [42], which can naturally unify early-Universe inflation with late-time cosmic acceleration while altering galactic dynamics; Moffat's scalar-tensor-vector gravity (STVG/MOG) [40], which modifies gravity via additional dynamical fields; retarded gravity models [49], which demonstrate that the finite propagation speed of gravitational effects over large galactic scales can mimic dark matter halos; and Hyperconical modified gravity (HMG) [41], which derives relativistic MOND-like phenomenology from the stereographic projection of higher-dimensional spacetimes.

These insights hint that gravitational anomalies may be thermodynamic and informational in origin. Motivated by this, we introduce two foundational principles. First, the time-energy equivalence principle posits that all forms of energy are manifestations of a single, underlying phase field whose value represents the local rate of quantum evolution. Second, the law of entropy equilibrium states that the Universe maintains a constant phase space, dynamically balancing the disordering drive of heat against the ordering drive of structure formation. From these two foundational principles, this paper constructs a macroscopic theory of emergent gravity.

This paper is organized as follows. Sections 2–4 introduce the thermodynamic axioms of the framework, proposing the time-energy equivalence principle and the law of entropy equilibrium to model the spacetime vacuum as a $U(1)$ wave-mechanical medium partitioned into coherent structure and chaotic fluctuations.

Sections 5 and 6 construct the formal geometry, deriving modified Einstein field equations, equations of motion, and an effective metric directly from the wave-mechanical interference between localized matter and the vacuum.

Section 7 applies this formalism to astrophysics, showing how the framework naturally recovers the MOND interpolating function for galaxies. It also addresses cluster mass anomalies—including the Bullet Cluster—suggesting a resolution via holographic equipartition and shock-induced thermodynamic phase transitions.

Sections 8–10 extend the framework to cosmology. By evaluating topological limits, we derive a theoretical matter density and a modified Friedmann equation. This approach substitutes the static cosmological constant with an evolving vacuum reservoir, offering a potential resolution to the cosmological coincidence problem.

Finally, section 11 proposes a parameter-free falsifiability criterion, predicting an anti-correlation between RAR residuals and local kinematic phase coherence. Future directions are outlined in section 12, followed by conclusions in section 13.

2. The time-energy equivalence principle

This theory is built upon the foundational idea that gravity is not a fundamental force, but an emergent thermodynamic phenomenon. To build this case, we first recall the analogy proposed by Jacobson [29] in his derivation of the Einstein field equations from thermodynamics.

He likened gravitational waves in spacetime to sound waves in a gas—collective, statistical phenomena rather than fundamental fields. Just as the equations of sound describe the thermodynamics of countless molecular collisions, the Einstein equations may describe the statistical behavior of a deeper medium. In this view, spacetime geometry is a macroscopic, emergent observable, with Einstein’s field equations serving as an equation of state valid only under local thermodynamic equilibrium. Inspired by this, we take the next step: identifying the underlying medium as a scalar field whose defining property is the very flow of time.

To define a new spacetime medium, we start by reexamining Einstein’s own equivalence principle. It states that the effects of gravity are locally indistinguishable from acceleration. From this, a universal connection emerges: acceleration causes time dilation (as seen in Special Relativity); gravity, being equivalent to acceleration, must therefore also cause time dilation; mass is the source of gravity, so it is fundamentally a source of time dilation; and radiation, carrying energy and momentum, also gravitates and is therefore also a source of time dilation.

This motivates a possible unifying interpretation: acceleration, mass, and radiation are all linked by their common ability to alter the local rate of time. This theory elevates this connection to a foundational principle: *The presence of any form of energy density within a region of spacetime is physically indistinguishable and equivalent to a localized slowing of the rate of time in that region.*

To formalize this principle mathematically, we define spacetime medium, as the most fundamental topological structure possible: a single, dimensionless quantum phase:

$$\phi(x^\mu) \in [0, 2\pi). \quad (1)$$

Because phase is inherently periodic, the fundamental state space of the Universe is a one-dimensional circle defined by the $U(1)$ gauge symmetry [47]. The macroscopic perception of time is simply the continuous evolution of this quantum phase. We call this medium the Timeflow field. The goal of this framework is not to propose a microscopic theory of quantum gravity, but to derive the macroscopic cosmological observables from the thermodynamics of the spacetime. Therefore, we treat the Timeflow field as an effective, macroscopic continuous medium.

The temporal dynamics of this medium are captured by the spacetime gradient of the phase. Assuming a mostly-plus metric signature $(-, +, +, +)$, the gradient of the phase dictates a strictly time-like 4-vector. Because the continuous $U(1)$ phase ϕ is dimensionless, its covariant gradient $\nabla_\mu \phi$ inherently possesses geometric dimensions of $[\text{length}]^{-1}$. We therefore define the local invariant geometric wavenumber, k , strictly as the geometric magnitude of this 4-gradient:

$$k \equiv \sqrt{-g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi}. \quad (2)$$

Physically, this wavenumber k represents the spatial density of phase cycles. Because the field possesses a measurable geometric flow, it defines a preferred local thermodynamic rest frame. The normalized, dimensionless 4-velocity [14] of the phase evolution naturally follows as:

$$u^\mu \equiv \frac{-g^{\mu\nu} \nabla_\nu \phi}{k}. \quad (3)$$

To explicitly connect the geometric wavenumber to the macroscopic flow of time, we evaluate the rate of phase evolution experienced by an observer comoving with the field. The observer's physical 4-velocity is defined as $V^\mu = c u^\mu$. The temporal frequency ω , the rate at which the phase evolves with respect to the observer's invariant proper time τ , is given by the directional derivative along the worldline:

$$\omega \equiv \frac{d\phi}{d\tau} = V^\mu \nabla_\mu \phi. \quad (4)$$

Substituting the normalized 4-velocity equation (3), we obtain:

$$\omega = c \left(\frac{-g^{\mu\nu} \nabla_\nu \phi}{k} \right) \nabla_\mu \phi = c \frac{k^2}{k} = ck. \quad (5)$$

This relation defines a kinematic identity arising from the normalization of the phase gradient. It reflects the fact that the scalar field defines a preferred local rest frame, with the phase gradient setting both the temporal direction and the intrinsic rate of evolution. Such constructions are well known in scalar-field theories with non-canonical kinetic structure, including k-essence [3] models and superfluid vacuum approaches [8], where the gradient of a scalar field defines an effective 4-velocity and a local clock for the medium.

Consequently, the proper time interval required for the phase to advance by $d\phi$ is:

$$d\tau = \frac{d\phi}{\omega} = \frac{d\phi}{ck}. \quad (6)$$

In this sense, the Timeflow field behaves as an effective relativistic medium whose internal phase evolution provides a physical realization of proper time.

Based on the time-energy equivalence principle, the invariant frequency ω of the field must be directly proportional to the local concentration of energy. This establishes a macroscopic thermodynamic parallel to the fundamental Planck–Einstein relation ($E = \hbar\omega$) [16], equating the structural energy of spacetime with the temporal frequency of its phase evolution.

To ensure this coupling is universally valid for all forms of energy, including massless radiation whose stress–energy trace vanishes, we couple the Timeflow frequency to the effective energy density ρ_{eff} measured by an observer comoving with the phase field:

$$\rho_{\text{eff}} \equiv T_{\mu\nu} u^\mu u^\nu. \quad (7)$$

Therefore, a high local energy density (ρ_{eff}) induces a proportionally high temporal frequency (ω). Because ω resides in the denominator of the proper time relation ($d\tau \propto 1/\omega$), this frequency physically manifests as the macroscopic slowing of proper time, natively reproducing the phenomenon of gravitational time dilation. From this perspective, all energy in the Universe is ultimately a manifestation of a single conserved quantity: the rate at which the quantum state of the Universe evolves.

3. The law of entropy equilibrium

A significant departure of this framework from standard General Relativity lies in the treatment of global energy conservation. In GR, the lack of a global timelike Killing vector in dynamic, expanding spacetime leads to the non-conservation of global energy [58]. However, the First Law of thermodynamics [12] forbids the creation or destruction of energy, and the quantum mechanical requirement for a unitary phase space [19] implies the fundamental preservation of information. Furthermore, the Bekenstein bound [7] demonstrates that the total informational capacity of any given region of spacetime is finite and strictly limited. If Timeflow is fundamentally a thermodynamic field, then its macroscopic dynamics should respect a global conservation principle of this kind. Therefore, we postulate that the Universe is an isolated thermodynamic system where total information is conserved.

According to the First Law, the change in the total internal energy of this closed system is strictly zero ($dE_{\text{total}} = 0$). The First Law is thus expressed as a dynamic equilibrium:

$$dE_{\text{total}} = \delta Q + \delta W = 0 \implies \delta Q = -\delta W. \quad (8)$$

This implies that over any period of cosmic evolution, the total thermal heat absorbed or redistributed within the vacuum (δQ) must exactly balance the total macroscopic work done to create and maintain gravitational structure (δW). If the Timeflow field universally satisfies this First Law, the Second Law of thermodynamics can be satisfied only if the underlying thermodynamic drives of heat and work are perfectly balanced.

The classical Second Law is experimentally valid for systems in which gravitational effects are negligible. In such contexts, energy flows primarily through heat and radiation, and the law correctly describes the tendency of energy to spread, equilibrate, and become less available for work. However, as emphasized by Roger Penrose [45], gravity fundamentally alters the entropy dynamics of the Universe. Unlike purely thermal processes, gravitational interaction concentrates energy, reduces local entropy, and creates highly ordered configurations. We therefore model the spacetime medium as governed by two complementary thermodynamic tendencies: a disordering drive associated with thermal incoherence, and an ordering drive associated with the formation of coherent structure:

1. Drive for order ($I_{\text{structure}} > 0$): This denotes the portion of the field configuration that is organized into coherent macroscopic structure. We interpret the energetic cost of creating and maintaining this ordered sector as thermodynamic work, W . Following the informational interpretation of negentropy due to Brillouin [9], we associate this ordering tendency with an increase in structural information:

$$\delta W = T dI_{\text{structure}}, \quad (9)$$

2. Drive for chaos ($S_{\text{chaos}} > 0$): This denotes the incoherent, thermally disordered sector of the field. It dictates the universal tendency toward maximum disorder, manifesting as a persistent entropic pressure driven by the field's incoherent zero-point oscillations. We associate its energy transfer with heat, Q , and describe it by the Clausius relation

$$\delta Q = T dS_{\text{chaos}}, \quad (10)$$

where S_{chaos} measures the entropic occupation of the unconstrained microscopic phase space.

To formalize these thermodynamic drives, we first specify the microscopic state space. Because the Timeflow field equation (1) is a compact phase variable with $U(1)$ symmetry, its microscopic configurations are defined on the manifold S^1 . The total accessible phase volume is therefore finite:

$$C_{\text{total}} = \oint_{S^1} d\phi = 2\pi. \quad (11)$$

This compactness implies that the local informational entropy of the field is bounded. In this sense, the quantity 2π is the finite geometric measure of the underlying microstate space from which the entropy is constructed.

We now identify the local thermodynamic entropy of the spacetime medium with the informational entropy of the phase distribution using the Boltzmann–Shannon framework [51]. At each macroscopic event x^μ , the local microstate of the field is described by a probability density $p(\phi)$ over the compact phase space, normalized as

$$\int_0^{2\pi} p(\phi) d\phi = 1. \quad (12)$$

The corresponding local entropy functional is

$$S[\phi] = -k_B \int_0^{2\pi} p(\phi) \ln p(\phi) d\phi. \quad (13)$$

By Jaynes' principle of maximum entropy [30], the physically realized macrostates are those selected by constraints on this distribution. In the present framework, those constraints are imposed by the local geometric wavenumber of the field, bounded between the infrared floor k_0 and the ultraviolet cutoff k_{max} .

In the incoherent vacuum state, no localized macroscopic constraint selects any preferred phase. Subject to the normalization constraint equation (12), and no further conditions on the phase, the entropy-maximizing distribution on the compact manifold S^1 is therefore uniquely the uniform distribution,

$$p(\phi) = \frac{1}{2\pi}. \quad (14)$$

This state represents the maximally disordered phase configuration of the field and corresponds to the infrared limit $k \rightarrow k_0$. Its entropy is

$$S_{\max} = -k_B \int_0^{2\pi} \frac{1}{2\pi} \ln\left(\frac{1}{2\pi}\right) d\phi = k_B \ln(2\pi). \quad (15)$$

In this limit, the field carries no structural information,

$$I_{\text{structure}} = 0, \quad (16)$$

and fully occupies the available microscopic phase space.

Conversely, perfect structural order corresponds to complete phase localization. In continuous probability theory, this idealized limit is represented by a Dirac delta distribution [19],

$$p(\phi) = \delta(\phi - \phi_0), \quad (17)$$

for which the differential entropy is formally divergent ($S \rightarrow -\infty$), reflecting infinite localization in phase space.

This singular behavior is unphysical and signals the breakdown of the continuum description at arbitrarily small scales. We therefore regularize the ordered limit by introducing a finite ultraviolet cutoff set by the minimal physically resolvable scale of the spacetime medium. In the present framework, this corresponds to the saturation of the geometric gradient at

$$k_{\max} = \frac{2\pi}{l_p}, \quad (18)$$

where $l_p \equiv \sqrt{\hbar G/c^3}$ is the Planck length, defining the fundamental quantum resolution scale of spacetime. This sets a minimum resolvable phase width $\Delta\phi_{\min} \sim 2\pi/k_{\max}$.

The coherent state is thus more properly described by a sharply localized but finite distribution of width $\Delta\phi_{\min}$, rather than an exact delta function. Its entropy is correspondingly finite and approaches a minimal value, which we adopt as the reference zero of structural uncertainty:

$$S[\phi] \rightarrow S_{\min} \equiv 0. \quad (19)$$

We then define the structural information of a macroscopic field configuration as the entropic deficit relative to the maximally disordered vacuum state:

$$I_{\text{structure}} \equiv S_{\max} - S[\phi]. \quad (20)$$

Therefore, as the field condenses from the thermal background (k_0) toward the Planck-scale limit ($k_{\max}$), the probability distribution narrows to its physical minimum, and the structural information saturates the fundamental capacity of the field ($I_{\text{structure}} \rightarrow S_{\max}$). This establishes that the invariant topological constant equation (11) is continuously partitioned between the field's two thermodynamic drives. Defining the chaotic entropy as

$$S_{\text{chaos}} \equiv S[\phi], \quad (21)$$

we obtain the exact partition:

$$S_{\max} = S_{\text{chaos}} + I_{\text{structure}}. \quad (22)$$

This relation is to be understood locally at each spacetime event, while the global conservation law emerges from integrating this balance over the entire spacetime manifold.

To quantify this dynamic partition, we define the equilibrium function, μ , as the thermodynamic efficiency of the local spacetime medium. Macroscopically, it represents the ratio of the structural coherent work to the total kinematic energy of the field:

$$\mu \equiv \frac{W}{E_{\text{total}}} = \frac{W}{W+Q}. \quad (23)$$

Informationally, this efficiency μ maps to the fraction of the universal intensive capacity that has been successfully phase-locked into structural order, cleanly interpolating between the boundary limits of k_0 ($\mu \rightarrow 0$) and k_{max} ($\mu \rightarrow 1$):

$$\mu = \frac{I_{\text{structure}}}{S_{\text{max}}} = \frac{I_{\text{structure}}}{I_{\text{structure}} + S_{\text{chaos}}}. \quad (24)$$

Because this topological boundary is fixed, the internal informational state of the Universe operates as a perfect zero-sum thermodynamic system. Any localized intensive increase in structural order ($\delta I_{\text{structure}} > 0$) is strictly balanced by an equal and opposite depletion of local chaos ($\delta S_{\text{chaos}} < 0$):

$$dS_{\text{chaos}} + dI_{\text{structure}} = 0. \quad (25)$$

This relation is the informational core of the present framework: order and disorder do not evolve independently, but exchange capacity within a fixed thermodynamic budget.

Therefore, to satisfy the First Law of thermodynamics and the kinematic boundaries of the phase space, we propose a new fundamental principle: *The total informational capacity of a spacetime region is conserved and dynamically balanced by an opposing process of structuring and dissipation, such that the sum of the thermal entropy and structural information remains invariant.*

This principle suggests an explanation of why coherent energy configurations have an inherent tendency to evolve towards greater complexity, progressing through a hierarchy of emergent structures. In this picture, the growth of complexity is a manifestation of the balance condition, where increases in local order are compensated by corresponding increases in surrounding chaos, preserving the total informational capacity.

The law of entropy equilibrium is thus stated as an equality between the thermodynamic entropy of the Universe, S_{chaos} , and the total information encoded in its gravitational structure, $I_{\text{structure}}$. This directly implies the classical Second Law of thermodynamics as an emergent consequence of structure formation. The observed increase in thermodynamic entropy is thus driven by the continuous writing of information into the cosmic web by gravity.

4. Wave-mechanics of spacetime

To connect the thermodynamic description above to spacetime dynamics, we now express the local energetic state of the medium in terms of the invariant phase gradient. The central object is the second moment of the covariant phase gradient, which encodes both coherent structure and incoherent fluctuations within a single macroscopic quantity.

In this framework, the energy of the spacetime medium is dictated by the gradient of its phase. We define the effective macroscopic wavenumber, k_{total} , as the invariant expectation value of the squared covariant phase gradient across the ensemble of microstates:

$$k_{\text{total}}^2 \equiv -g^{\mu\nu} \langle (\nabla_{\mu} \phi_{\text{total}}) (\nabla_{\nu} \phi_{\text{total}}) \rangle. \quad (26)$$

Then we adopt a coarse-grained decomposition into an ordered mean component and stochastic fluctuations:

$$\phi_{\text{total}} = \phi_{\text{order}} + \phi_{\text{chaos}}, \quad (27)$$

or equivalently,

$$\nabla_{\mu} \phi_{\text{total}} = \nabla_{\mu} \phi_{\text{order}} + \nabla_{\mu} \phi_{\text{chaos}}. \quad (28)$$

The linear form of this decomposition is forced by the First Law of thermodynamics in equilibrium form equation (8), combined with the differential identities δW equation (9) and δQ equation (10): a unit transfer between the two sectors must produce equal and opposite changes in $I_{\text{structure}}$ and S_{chaos} ,

independent of the current field configuration. A non-additive decomposition would introduce a configuration-dependent exchange rate between the sectors, breaking $\delta Q = -\delta W$ as an exact differential relation under the framework's local temperature T . The linear gradient decomposition is therefore the unique coarse-graining consistent with the First Law.

Here $\nabla_\mu \phi_{\text{order}}$ denotes the coherent, phase-locked component of the field, which we associate macroscopically with localized matter and radiation, while $\nabla_\mu \phi_{\text{chaos}}$ denotes the incoherent fluctuating background associated with the thermal vacuum. This decomposition is understood as a statistical coarse-graining of the field, analogous to the separation of an order parameter from fluctuations in many-body and scalar-field systems [16, 60], as well as in Bose–Einstein condensates [48], where a macroscopic coherent phase emerges from an underlying quantum field with thermal and quantum fluctuations.

Substituting this superposition into our invariant wavenumber yields the macroscopic second moment of the phase gradient:

$$k_{\text{total}}^2 = -g^{\mu\nu} \langle (\nabla_\mu \phi_{\text{order}} + \nabla_\mu \phi_{\text{chaos}}) (\nabla_\nu \phi_{\text{order}} + \nabla_\nu \phi_{\text{chaos}}) \rangle. \quad (29)$$

Expanding this quadratic form reveals three distinct thermodynamic terms:

$$k_{\text{total}}^2 = \langle -g^{\mu\nu} \nabla_\mu \phi_{\text{order}} \nabla_\nu \phi_{\text{order}} \rangle + \langle -g^{\mu\nu} \nabla_\mu \phi_{\text{chaos}} \nabla_\nu \phi_{\text{chaos}} \rangle + 2 \langle -g^{\mu\nu} \nabla_\mu \phi_{\text{order}} \nabla_\nu \phi_{\text{chaos}} \rangle. \quad (30)$$

Applying our thermodynamic constraints to this bounded phase space, we can physically identify these kinematic components. The expectation value of the coherent signal represents the geometric work capacity of the medium, manifesting macroscopically as all localized, phase-locked energy of bare matter and radiation:

$$k_{\text{order}}^2 = \langle -g^{\mu\nu} \nabla_\mu \phi_{\text{order}} \nabla_\nu \phi_{\text{order}} \rangle. \quad (31)$$

The variance of the stochastic noise represents the irreducible thermal heat of the bare spacetime vacuum:

$$k_0^2 = \langle -g^{\mu\nu} \nabla_\mu \phi_{\text{chaos}} \nabla_\nu \phi_{\text{chaos}} \rangle. \quad (32)$$

The final term represents the wave-mechanical interference between the coherent matter signal and the chaotic vacuum background. The presence of localized, coherent matter acts as a non-linear driving force, inducing local phase-locking in the surrounding chaotic background.

In continuous field mechanics, the expectation value of two partially phase-locked wave states evaluates to the product of their root-mean-square magnitudes and their Pearson correlation coefficient (ρ). Because the massive coherent soliton acts as the dominant non-linear driver, the chaotic thermal background minimizes the local interaction energy by tending toward partial phase alignment with the coherent source. A negative correlation ($\rho < 0$), representing systemic destructive interference, would require the thermal vacuum to spontaneously organize an active, anti-aligned gradient. Since a random thermal bath cannot spontaneously generate sustained anti-aligned work without an external driver, the correlation coefficient is can be reasonably expected to lie within the non-negative domain:

$$\rho \in [0, 1]. \quad (33)$$

In the geometric representation of the $U(1)$ manifold, this coefficient maps identically to the cosine of the phase alignment angle θ between the coherent matter signal ($\nabla_\mu \phi_{\text{order}}$) and the chaotic vacuum fluctuations ($\nabla_\mu \phi_{\text{chaos}}$):

$$\langle -g^{\mu\nu} \nabla_\mu \phi_{\text{order}} \nabla_\nu \phi_{\text{chaos}} \rangle = k_{\text{order}} k_0 \cos \theta. \quad (34)$$

Consequently, the physical boundaries for the phase alignment are:

$$\theta \in \left[0, \frac{\pi}{2}\right]. \quad (35)$$

Substituting these definitions, the total kinematic capacity of the local spacetime medium recovers the statistical variance sum law:

$$k_{\text{total}}^2 = k_{\text{order}}^2 + k_0^2 + 2k_{\text{order}} k_0 \cos \theta. \quad (36)$$

To determine the true thermodynamic partition of this total capacity, we calculate the geometric projection of the unified flow onto its respective constituent states via the metric inner product.

The structural work capacity (W) is the projection of the total kinematic footprint strictly onto the ordered macroscopic state ($\nabla_\mu \phi_{\text{order}}$):

$$\langle -g^{\mu\nu} (\nabla_\mu \phi_{\text{order}} + \nabla_\mu \phi_{\text{chaos}}) \nabla_\nu \phi_{\text{order}} \rangle = k_{\text{order}}^2 + k_{\text{order}} k_0 \cos \theta. \quad (37)$$

Because energy is proportional to the wavenumber squared, this represents the geometric work capacity:

$$W \propto k_{\text{order}}^2 + k_{\text{order}} k_0 \cos \theta. \quad (38)$$

Conversely, the chaotic heat (Q) is the projection of the total kinematic footprint onto the random thermal background ($\nabla_\mu \phi_{\text{chaos}}$):

$$\langle -g^{\mu\nu} (\nabla_\mu \phi_{\text{order}} + \nabla_\mu \phi_{\text{chaos}}) \nabla_\nu \phi_{\text{chaos}} \rangle = k_0^2 + k_{\text{order}} k_0 \cos \theta, \quad (39)$$

$$Q \propto k_0^2 + k_{\text{order}} k_0 \cos \theta. \quad (40)$$

This geometric projection inherently divides the interference cross-term ($2k_{\text{order}} k_0 \cos \theta$) equally. The shared interaction capacity mathematically belongs half to the structured capacity doing the work, and half to the thermal background exerting the pressure.

Therefore we define the equilibrium function μ equation (23), as the wave-mechanical efficiency of the local field. Macroscopically, it represents the ratio of the structural work capacity to the total energy footprint. Because the conversion factors between wavenumber and energy ($E = \hbar ck$) cancel out in the ratio, μ evaluates exactly as:

$$\mu = \frac{W}{W+Q} = \frac{k_{\text{order}}^2 + k_{\text{order}} k_0 \cos \theta}{k_{\text{order}}^2 + k_0^2 + 2k_{\text{order}} k_0 \cos \theta} = \frac{k_{\text{order}}^2 + k_{\text{order}} k_0 \cos \theta}{k_{\text{total}}^2}. \quad (41)$$

When matter is dynamically cold and highly ordered, constructive interference dominates ($\mu \rightarrow 1$). The informational capacity is fully saturated by structural information ($I_{\text{structure}} \rightarrow S_{\text{max}}$), and the field's energy is entirely available to bend macroscopic geometry. Conversely, in regions of deep void or high thermal chaos, phase decoherence drives the interference term to zero ($\mu \rightarrow 0$). The structural information evaporates, and the capacity dissolves entirely into the chaotic thermal background ($S_{\text{chaos}} \rightarrow S_{\text{max}}$).

5. Field equations

Having derived the equilibrium function μ equation (41) from the wave-mechanics of the $U(1)$ phase space equation (1), we now generalize this result to a covariant macroscopic theory. We achieve this by applying it to the thermodynamic argument of Jacobson [29], which demonstrated that the Einstein field equation is an equation of state for the spacetime medium.

Jacobson's derivation considers an arbitrary point in spacetime and the local Rindler horizon perceived by an accelerating observer. The Clausius relation, $\delta Q = T dS$, is applied to this local causal boundary.

The heat flux δQ is identified with the energy of matter crossing the horizon, as given by the flux of the matter stress–energy tensor, $T_{\mu\nu}^{(m)}$. The temperature T is the local Unruh [53] temperature of the accelerating frame. The entropy change dS is assumed to be proportional to the change in the horizon's area, δA , with the standard gravitational constant G dictating the fundamental Planck-scale proportionality:

$$dS_{\text{BH}} = \frac{k_B c^3}{4\hbar G} \delta A. \quad (42)$$

Demanding that the Clausius relation ($\delta Q = T dS_{\text{BH}}$) holds for all local Rindler observers fundamentally restricts the emergent geometry of spacetime, forcing the metric to obey the standard Einstein field equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{(m)}. \quad (43)$$

This theory modifies this derivation at the final step by directly applying the thermodynamic partition of the Timeflow field to the energy flux.

In this framework, matter is a localized, phase-locked structural excitation of the $U(1)$ vacuum. Because the efficiency of the vacuum is strictly bounded by μ equation (41), the matter tensor ($T_{\mu\nu}^{(m)}$) represents only the coherent structural work (W) equation (38). The total effective thermodynamic load (δQ_{eff}) imposed on the causal horizon must also include the chaotic thermal cost (Q) equation (40) required by the vacuum to maintain that structure.

Therefore, the matter flux is only a fraction of the total energetic footprint. Following Jacobson, the classical heat flux is the integral of the matter stress–energy tensor contracted with the horizon’s null generators, k^μ . In this framework, the total effective thermodynamic flux (δQ_{total}) that the space–time geometry must dynamically balance is this matter flux scaled by the inverse of our efficiency factor equation (41):

$$\delta Q_{\text{total}} = \frac{\delta W}{\mu} = \frac{1}{\mu} \int T_{\mu\nu}^{(m)} k^\mu k^\nu d\Sigma d\lambda. \quad (44)$$

Applying the First Law to the local horizon, the emergent geometry responds to this total partitioned flux. Equating the total thermodynamic heat to the classical geometric capacity of the horizon yields the modified Clausius relation:

$$\frac{1}{\mu} \int T_{\mu\nu}^{(m)} k^\mu k^\nu d\Sigma d\lambda = T dS_{\text{BH}}. \quad (45)$$

Requiring this exact relation to hold for all local Rindler observers and all null vectors k^μ implies that the integrals must be identically equal, leading to a modified Einstein field equation of the form:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left(\frac{T_{\mu\nu}^{(m)}}{\mu} \right). \quad (46)$$

The modification arises from the dynamic local wave-mechanical state of the vacuum (μ) equation (41), which dictates what fraction of the energy flux successfully generates spatial curvature.

A fundamental requirement for any metric theory of gravity is the local conservation of energy and momentum. In standard General Relativity, this is guaranteed by the contracted Bianchi identity [14], $\nabla^\mu G_{\mu\nu} = 0$. To recover the strictly conserved standard form of GR, we must establish the total macroscopic stress–energy tensor of the spacetime medium.

Macroscopically, the total kinematic capacity of the field is fundamentally defined by the statistical expectation value of its phase gradients equation (26). We define this total effective stress–energy tensor as:

$$T_{\mu\nu}^{\text{total}} = \mathcal{C} \langle (\nabla_\mu \phi_{\text{total}}) (\nabla_\nu \phi_{\text{total}}) \rangle. \quad (47)$$

where $\mathcal{C}$ is a fundamental dimensional constant.

The constant $\mathcal{C}$ must emerge directly from the fundamental symmetries and limits of the medium. We determine its exact composition through dimensional analysis. The 00-component of the stress–energy tensor represents energy density, which in SI units has dimensions of Joules per cubic meter ($\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}$). Under our geometric definition, the squared invariant wavenumber k^2 equation (2) carries geometric dimensions of inverse square meters (m^{-2}). Dimensional equivalence therefore requires:

$$[\mathcal{C}] = \frac{[T_{00}]}{[k^2]} = \frac{\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}}{\text{m}^{-2}} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}. \quad (48)$$

Therefore, $\mathcal{C}$ must represent a fundamental unit of force (Newtons). The only fundamental constants available that govern spacetime mechanics are the speed of light (c) and the gravitational constant (G). To generate units of force using only c and G , there is exactly one unique algebraic combination:

$$\frac{c^4}{G} = \frac{\text{m}^4 \cdot \text{s}^{-4}}{\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}. \quad (49)$$

This combination is recognized as the fundamental Planck force. The dimensional analysis fixes the normalization only up to a dimensionless constant, implying

$$\mathcal{C} \sim \frac{c^4}{G}. \quad (50)$$

To determine the exact coefficient, we require that the macroscopic field equation recover the standard Newtonian limit in the weak-field, non-relativistic regime. In this limit, the Einstein tensor satisfies

$$G_{00} \simeq \frac{2}{c^2} \nabla^2 \Phi, \quad (51)$$

while the matter tensor reduces to

$$T_{00} \simeq \rho c^2. \quad (52)$$

Demanding consistency with Poisson's equation,

$$\nabla^2 \Phi = 4\pi G \rho, \quad (53)$$

fixes the coupling to

$$\mathcal{C} = \frac{c^4}{8\pi G}. \quad (54)$$

By substituting our constant equation (54) alongside our phase-gradient tensor definition equation (47) into the standard Einstein field equations, we yield:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{\text{total}} = \frac{8\pi G}{c^4} \left[\left(\frac{c^4}{8\pi G} \right) \langle (\nabla_\mu \phi_{\text{total}}) (\nabla_\nu \phi_{\text{total}}) \rangle \right]. \quad (55)$$

The fundamental force of the $U(1)$ field in the numerator cancels the Einstein geometric conversion factor in the denominator. The equation reduces to a compact geometric relation:

$$G_{\mu\nu} = \langle (\nabla_\mu \phi_{\text{total}}) (\nabla_\nu \phi_{\text{total}}) \rangle. \quad (56)$$

This suggests that, in the present framework, macroscopic spacetime curvature is sourced directly by the ensemble-averaged phase-gradient structure of the $U(1)$ field.

We can now expand this geometric identity into its wave-mechanical states. By substituting the coarse-grained decomposition of the phase field ($\nabla_\mu \phi_{\text{total}} = \nabla_\mu \phi_{\text{order}} + \nabla_\mu \phi_{\text{chaos}}$) into equation (56), we expand the expectation value of the quadratic form. Ensuring the interference cross-term strictly respects the symmetry of the Einstein tensor, we yield:

$$G_{\mu\nu} = \langle \nabla_\mu \phi_{\text{order}} \nabla_\nu \phi_{\text{order}} \rangle + \langle \nabla_\mu \phi_{\text{chaos}} \nabla_\nu \phi_{\text{chaos}} \rangle + 2 \langle \nabla_{(\mu} \phi_{\text{order}} \nabla_{\nu)} \phi_{\text{chaos}} \rangle. \quad (57)$$

To map these topological gradients to macroscopic physical observables, we apply the geometric kinematic relationship defined by our 4-velocity equation (3). Evaluating the trace of the phase kinematics allows us to identify the distinct geometric components of the vacuum corresponding to the total kinematic phase capacity equation (36):

Matter is represented by the expectation value of the coherent macroscopic signal, yielding a directed, phase-locked flow. This represents the structured work of the medium, manifesting as the geometric well of a pressureless fluid governed by its 4-velocity (u_μ):

$$\langle \nabla_\mu \phi_{\text{order}} \nabla_\nu \phi_{\text{order}} \rangle = k_{\text{order}}^2 u_\mu u_\nu. \quad (58)$$

The thermal background noise is represented by the stochastic zero-point fluctuations of the phase field that are locally isotropic and uncorrelated. Isotropy requires this tensor to be proportional to $g_{\mu\nu}$, and the coefficient is fixed by the trace condition

$$-g^{\mu\nu} \langle \nabla_\mu \phi_{\text{chaos}} \nabla_\nu \phi_{\text{chaos}} \rangle = k_0^2, \quad (59)$$

which yields the factor 1/4 in four spacetime dimensions. Enforcing the trace boundary for the chaotic heat, the vacuum variance assumes the form of an intrinsic negative pressure:

$$\langle \nabla_\mu \phi_{\text{chaos}} \nabla_\nu \phi_{\text{chaos}} \rangle = -\frac{1}{4} k_0^2 g_{\mu\nu}. \quad (60)$$

The cross-correlation between the bare mass and the thermal background ($2k_{\text{order}}k_0 \cos \theta$) manifests as a phase-locked interference halo. Because this structural negentropy is strictly anchored to the coherent mass, it shares the macroscopic kinematics of the bare matter (u_μ), yielding:

$$2 \langle \nabla_{(\mu} \phi_{\text{order}} \nabla_{\nu)} \phi_{\text{chaos}} \rangle = (2k_{\text{order}}k_0 \cos \theta) u_\mu u_\nu. \quad (61)$$

Substituting these wave-mechanical states yields the complete, expanded field equation of Timeflow Gravity:

$$G_{\mu\nu} = (k_{\text{order}}^2 + 2k_{\text{order}}k_0 \cos\theta) u_\mu u_\nu - \frac{1}{4}k_0^2 g_{\mu\nu}. \quad (62)$$

This result suggests that dark matter and dark energy effects could emerge from the same unified stress–energy tensor, governed entirely by the wave mechanics of the $U(1)$ Timeflow field. In dynamically hot, pressure-supported regions or vast cosmic voids, random thermal motion causes severe phase decoherence ($\cos\theta \rightarrow 0$). The phase-locked interference term vanishes, leaving the massive bare vacuum heat (k_0^2) to dominate the spacetime geometry. This behaves effectively as a repulsive cosmological background, which standard cosmology identifies as dark energy. Conversely, in rotationally supported galaxies, the ordered, low-entropy motion of baryons maintains a highly coherent phase relationship with the vacuum ($\cos\theta \rightarrow 1$). This strict correlation phase-locks the chaotic vacuum into a localized, non-linear halo that amplifies the gravitational field, which may correspond to what is observationally attributed to dark matter.

6. Equation of motion and metric

If matter represents a localized coherent state with a high geometric wavenumber k , and the surrounding vacuum represents an incoherent state of a lower baseline wavenumber k_0 , a steep entropic gradient naturally forms between them. In standard thermodynamics, a system experiences an emergent statistical force driving it toward equilibrium—minimizing energetic potentials and maximizing entropy. In the present framework, this gradient manifests as a spatial variation in the macroscopic density of phase cycles, $\nabla_\mu k$. In this view, macroscopic gravity is not an intrinsic fundamental interaction, but rather this emergent entropic force.

To derive the classical equation of motion for a test mass m within this thermodynamic medium, we evaluate its local rest energy. According to equation (6), the proper time $d\tau$ experienced by a test mass is inversely proportional to the local macroscopic wavenumber of the field. We define the dimensionless Timeflow factor $\mathcal{T}(x^\mu)$ as the ratio of the vacuum ground-state wavenumber to the local wavenumber:

$$\mathcal{T}(x^\mu) \equiv \frac{k_0}{k(x^\mu)}. \quad (63)$$

Because the dynamics of matter are governed by the local phase evolution rate $\omega = d\phi/d\tau$ equation (5), which directly defines the flow of proper time equation (6), the effective rest energy of a test particle is modulated by the Timeflow factor equation (63). In the absence of gradients (deep vacuum), the particle possesses its standard rest energy $E = mc^2$. In the presence of a macroscopic phase gradient, this energy is locally rescaled:

$$E(x^\mu) = mc^2 \mathcal{T}(x^\mu) = mc^2 \left(\frac{k_0}{k(x^\mu)} \right). \quad (64)$$

To determine the particle trajectory, we construct the action. In relativistic mechanics, the action of a free particle is proportional to the proper time along its worldline. In the present framework, this proper time is modulated by the local field by the factor equation (63). The action therefore generalizes to

$$S = -mc^2 \int \mathcal{T}(x^\mu) d\tau. \quad (65)$$

This can be interpreted as motion in an effective spacetime geometry where the physical proper time is rescaled:

$$d\tau_{\text{eff}} = \mathcal{T}(x^\mu) d\tau. \quad (66)$$

Thus, the action takes the standard geodesic form

$$S = -mc^2 \int d\tau_{\text{eff}}, \quad (67)$$

demonstrating that particles follow geodesics of an emergent metric determined by the Timeflow field.

To make this explicit, we rewrite the action in coordinate time. Using

$$d\tau = \sqrt{-g_{\mu\nu}dx^\mu dx^\nu}/c,$$

the Lagrangian becomes

$$L = -mc^2\mathcal{T}(x)\sqrt{1 - \frac{v^2}{c^2}}. \quad (68)$$

In the non-relativistic limit ($v \ll c$), this expands to

$$L \approx -mc^2\mathcal{T}(x) + \frac{1}{2}m\mathcal{T}(x)v^2. \quad (69)$$

The first term acts as an effective potential, while the second modifies the inertial response. The force acting on the particle follows from the Euler–Lagrange equations ($\frac{d}{dt}\frac{\partial L}{\partial v_i} = \frac{\partial L}{\partial x_i}$):

$$\frac{d}{dt}(m\mathcal{T}v_i) = -mc^2\partial_i\mathcal{T} + \frac{1}{2}mv^2\partial_i\mathcal{T}. \quad (70)$$

Neglecting velocity-dependent corrections in the strict slow-motion limit, this reduces to:

$$\frac{d}{dt}(m\mathcal{T}v_i) \approx -mc^2\partial_i\mathcal{T}. \quad (71)$$

Dividing by the effective inertial mass ($m\mathcal{T}$) and assuming the temporal variation of $\mathcal{T}$ is negligible for local non-relativistic motion, this yields the classical acceleration:

$$a_i = -c^2\frac{\partial_i\mathcal{T}}{\mathcal{T}} = -c^2\partial_i\ln\mathcal{T}. \quad (72)$$

Promoting this to a covariant expression yields the four-acceleration:

$$a_\mu = -c^2\nabla_\mu\ln\mathcal{T}(x^\mu). \quad (73)$$

Thus, gravitational acceleration emerges as the gradient of the logarithm of the Timeflow factor. The logarithmic form arises because the effective rest energy depends multiplicatively on $\mathcal{T}$, so its gradient naturally produces $\nabla\ln\mathcal{T}$.

This reveals a deep thermodynamic connection to Boltzmann’s definition of entropy via the logarithm of microscopic states ($S \propto \ln W$). We define the classical gravitational potential $\Phi(x^\mu)$ exactly as the natural logarithm of the Timeflow factor equation (63). By recognizing that the macroscopic wavenumber of the field $k(x^\mu)$ is identically the total root-mean-square wavenumber derived via the wave-mechanical superposition (k_{total}) equation (36), the potential is naturally defined as:

$$\Phi(x^\mu) \equiv c^2\ln\left(\frac{k_0}{k_{\text{total}}(x^\mu)}\right), \quad (74)$$

$$a_\mu = -\nabla_\mu\Phi(x^\mu). \quad (75)$$

Because the local wavenumber near a mass is driven above the vacuum baseline ($k_{\text{total}} > k_0$), the ratio $\mathcal{T}$ is strictly less than unity. Consequently, the logarithm is negative, ensuring an attractive gravitational potential ($\Phi < 0$).

To formally map this emergent thermodynamic acceleration to the geometric framework of General Relativity, we define the spacetime metric $g_{\mu\nu}$ directly through this potential. Because proper time $d\tau$ equation (6) is dictated by the invariant macroscopic evolution of the field, the temporal component of the effective metric (g_{00}) is simply the square of the Timeflow factor equation (63). Assuming a mostly-plus signature $(-, +, +, +)$, this yields an exact exponential mapping:

$$g_{00} = -\left(\frac{k_0}{k_{\text{total}}(x^\mu)}\right)^2 = -\mathcal{T}(x^\mu)^2 = -e^{2\Phi(x^\mu)/c^2}. \quad (76)$$

Furthermore, to ensure that massless radiation correctly couples to the emergent metric, we must satisfy the invariant topological capacity of the $U(1)$ phase space equation (11). For a photon traversing

this thermodynamic medium, its phase-space volume must be strictly conserved. If the temporal dimension kinematically dilates ($k > k_0$), the spatial dimensions is expected to rescale isotropically to preserve the total trace of the effective thermodynamic flux. In isotropic coordinates, this boundary condition demands that the spatial curvature g_{ij} scales by an inverse conformal factor relative to the temporal metric:

$$g_{ij} = \delta_{ij} e^{-2\Phi(x^\mu)/c^2}. \quad (77)$$

Substituting these correlated temporal and spatial components yields the static, spherically symmetric line element for Timeflow Gravity in isotropic coordinates:

$$ds^2 = -e^{2\Phi(r)/c^2} c^2 dt^2 + e^{-2\Phi(r)/c^2} (dr^2 + r^2 d\Omega^2). \quad (78)$$

The resulting isotropic metric equation (78) yields an exponential geometry historically associated with the scalar-tensor framework proposed by Yilmaz [61]. This metric is known for its ability to satisfy the parameterized post-Newtonian (PPN) formalism [27]. A Taylor expansion of the temporal metric to order $(v/c)^4$ yields $g_{00} \approx -(1 + 2\Phi/c^2 + 2\Phi^2/c^4)$, yielding the PPN non-linearity parameter $\beta = 1$. Simultaneously, the expansion of the spatial metric yields $g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2)$, setting the spacetime curvature parameter $\gamma = 1$. Because this metric intrinsically yields $\beta = \gamma = 1$, it recovers the standard isotropic Schwarzschild expansion. This suggests that the Timeflow metric may be consistent with key precision tests of General Relativity, including the anomalous perihelion precession of Mercury and the Shapiro time delay of radiation.

Furthermore, due to the exponential structure of the metric equation (78), the theory naturally avoids the formation of Schwarzschild singularities [27]. However, a rigorous mathematical treatment of this extreme strong-field regime and its horizon thermodynamics falls outside the scope of the present paper and remains a subject for future work.

7. Galaxies and clusters dynamics

Having established the modified Einstein field equations from the thermodynamics of the Timeflow field, we now examine the weak-field, non-relativistic limit. We demonstrate how the full equilibrium function μ naturally yields MOND [35] for coherent systems, while its phase-dependence resolves the longstanding mass discrepancies in galaxy clusters.

We take the weak-field, non-relativistic trace of our previously derived modified Einstein Field equation equation (46). In this limit, the matter stress–energy tensor is dominated by its rest-mass density ($T_{00}^{(m)} \approx \rho_{\text{bare}} c^2$). Substituting the metric equation (78) into the modified field equation yields a non-linear Poisson equation:

$$\nabla \cdot (\mu (\nabla \Phi) \nabla \Phi) = 4\pi G \rho_{\text{bare}}. \quad (79)$$

This has the same mathematical structure as the Bekenstein–Milgrom AQUA formulation of MOND [5]. It shows that the spatial gradient of time is fundamentally modulated by the wave-mechanical efficiency of the vacuum (μ), inherently guaranteeing the conservation of momentum and angular momentum.

To connect the foundational thermodynamics of the spacetime medium to macroscopic observables, we must relate the entropic drives of the $U(1)$ field to measurable kinematic acceleration. As established by the law of entropy equilibrium equation (23), the total kinematic capacity of the field is partitioned into structural work and chaotic heat where the differential energy transfers are driven by the local temperature T of the medium equations (9) and (10).

Because macroscopic acceleration physically shifts the local causal horizon of an observer, it natively induces a thermal gradient in the quantum vacuum via the Unruh effect [53]. The temperature T experienced by a macroscopic structure undergoing coherent radial acceleration a is:

$$T_{\text{order}} = \frac{\hbar a}{2\pi k_B c}. \quad (80)$$

The conceptual link between the Unruh effect, the vacuum temperature, and the MOND acceleration scale a_0 builds upon foundational insights in modified dynamics. Milgrom [39] notably proposed that MOND inertia is inherently a vacuum effect, wherein an accelerated observer perceives a modified Unruh radiation temperature governed by a cosmic background acceleration a_0 . Expanding on this

thermodynamic connection, Darabi [18] demonstrated that the interplay between a local Rindler horizon and a cosmological horizon generates an Unruh-like effect, naturally encoding a_0 as a fundamental threshold in the effective dynamics. In our framework, we formally realize this relationship: the irreducible zero-point variance of the chaotic vacuum establishes a fundamental, universal baseline temperature bounded exactly by this critical acceleration scale a_0 :

$$T_0 = \frac{\hbar a_0}{2\pi k_B c}. \quad (81)$$

In the Euclidean formulation of Rindler spacetime, regularity at the horizon requires the Euclidean proper-time coordinate to be periodic with period [29, 54]

$$\Delta\tau_E = \frac{2\pi c}{a}. \quad (82)$$

In this framework, proper time is defined intrinsically by the phase evolution of the field, through the fundamental relation equation (6). Therefore, a periodicity in Euclidean proper time directly implies a fundamental periodicity of the phase field itself. The Euclidean cycle defines a closed orbit in phase evolution, and thus fixes the characteristic angular frequency of the field:

$$\omega \equiv \frac{2\pi}{\Delta\tau_E} = \frac{2\pi}{2\pi c/a} = \frac{a}{c}. \quad (83)$$

Thus, the Unruh effect implies a direct and exact correspondence between proper acceleration and the intrinsic phase evolution rate of the Timeflow field:

$$a = c\omega. \quad (84)$$

This establishes that macroscopic acceleration is not merely a kinematic quantity, but the geometric manifestation of the underlying phase frequency of the spacetime medium. Applying the same phase-acceleration correspondence to the vacuum baseline gives:

$$\omega_0 = \frac{a_0}{c}. \quad (85)$$

We now substitute these exact kinematic mappings directly into the wave-mechanical equilibrium function equation (41). Based on our definition of frequency equation (5) yields:

$$\mu(a, a_0, \theta) = \frac{\left(\frac{a}{c}\right)^2 + \left(\frac{a}{c}\right)\left(\frac{a_0}{c}\right)\cos\theta}{\left(\frac{a}{c}\right)^2 + \left(\frac{a_0}{c}\right)^2 + 2\left(\frac{a}{c}\right)\left(\frac{a_0}{c}\right)\cos\theta}. \quad (86)$$

Factoring out the shared invariant kinematic scaling factor $(1/c)$ from both the numerator and denominator, the speed of light cancels. The thermodynamic efficiency collapses into a macroscopic interpolating function governed entirely by acceleration and phase coherence:

$$\mu(a, a_0, \theta) = \frac{a^2 + aa_0 \cos\theta}{a^2 + a_0^2 + 2aa_0 \cos\theta}. \quad (87)$$

This function reveals that gravitational amplification is not a static rule, but a dynamic wave-mechanical response. In rotationally supported spiral galaxies, the motion of stars and gas is dynamically cold and highly ordered. The macroscopic phase field maintains a stable, synchronized alignment with the vacuum, meaning constructive interference is maximized ($\cos\theta \approx 1$). For these coherent galactic systems, the equilibrium function simplifies to:

$$\mu_{\text{galaxy}} \approx \frac{a^2 + aa_0}{a^2 + a_0^2 + 2aa_0} = \frac{a(a + a_0)}{(a + a_0)^2} = \frac{a}{a + a_0}. \quad (88)$$

In this coherent limit, the equilibrium function μ of equation (87) reduces exactly to the simple MOND interpolating function of Milgrom [35]. The shared symbol μ reflects that Milgrom's empirical interpolating function emerges as the coherent-limit form of the framework's thermodynamic equilibrium function.

We analyze this coherent response in the deep-MOND regime (the outer edges of galaxies), where the gravitational acceleration falls far below the vacuum baseline ($a \ll a_0$). In this limit, the a^2 terms become negligible, and the function is entirely dominated by the a_0 terms:

$$\mu_{\text{galaxy}}(a \ll a_0) \approx \frac{aa_0}{a_0^2} = \frac{a}{a_0}. \quad (89)$$

Because our thermodynamic equation of state dictates that the Newtonian acceleration (g_N) relates to the effective physical acceleration (a) via $g_N = a \cdot \mu$, we find:

$$g_N \approx a \left(\frac{a}{a_0} \right) = \frac{a^2}{a_0} \implies a = \sqrt{g_N a_0}. \quad (90)$$

This reproduces the standard MOND scaling relation. The gravitational acceleration transitions from an inverse-square law ($1/r^2$) to an inverse-linear law ($1/r$), naturally generating the flat rotation curves of spiral galaxies. So in this view, the MONDian amplification is the physical footprint of constructive phase interference between matter and the vacuum.

7.1. Galaxy clusters

While empirical MOND scaling relations successfully predict galactic rotation curves, they consistently underpredict the dynamical mass of massive galaxy clusters [23]. Timeflow Gravity offers a potential explanation for this discrepancy through the macroscopic thermodynamic partition of the field equations. In this framework, the observational difference between a galaxy and a cluster is the macroscopic manifestation of the phase coherence angle θ and the resulting wave-mechanical interference.

Unlike rotationally supported spiral galaxies, which maintain dynamically cold, low-entropy baryonic flows ($\cos\theta \approx 1$), galaxy clusters are pressure-supported systems characterized by immense virial velocity dispersions ($\sigma_v \sim 1000 \text{ km s}^{-1}$). The constituent galaxies and intracluster gas move in random, chaotic, intersecting trajectories. This extreme kinematic entropy causes rapid phase fluctuations, strongly suppressing the phase correlation between the localized matter gradient and the vacuum background.

We evaluate our expanded field equation (62) in this highly decoherent limit ($\cos\theta \rightarrow 0$). The wave-mechanical interference term ($2k_{\text{order}}k_0 \cos\theta$) vanishes. The macroscopic geometry collapses to:

$$G_{\mu\nu} = k_{\text{order}}^2 u_\mu u_\nu - \frac{1}{4} k_0^2 g_{\mu\nu}. \quad (91)$$

Without the constructive interference term to organize the vacuum into a structured spatial gradient, a traditional $1/r$ dark matter halo is suppressed. The localized effective dynamical mass is strictly reduced to the bare baryonic matter ($k_{\text{order}}^2 u_\mu u_\nu$). However, the macroscopic geometry is now heavily immersed in the irreducible thermal heat of the vacuum ($-\frac{1}{4}k_0^2 g_{\mu\nu}$), which acts as an isotropic pressure rather than a localized mass.

Because this chaotic heat is completely uncoupled from localized baryonic matter, its macroscopic influence must be evaluated geometrically. According to our expanded field equation (62), the uncoupled vacuum tensor is strictly proportional to the spacetime metric ($g_{\mu\nu}$). Since Riemannian metric compatibility demands $\nabla_\alpha g_{\mu\nu} = 0$, the chaotic vacuum is perfectly isotropic and devoid of internal spatial gradients ($\nabla_i \rho_{\text{vac}} = 0$).

The chaotic vacuum represents a state of maximum thermal entropy equation (15), so its degrees of freedom are fully saturated. According to the Holographic Principle [52], the thermodynamic capacity of such a saturated state cannot scale with its 3D volume; it is strictly bounded by its enveloping area. Consequently, the vacuum's macroscopic thermodynamic footprint is naturally modeled by an effective holographic contribution associated with the system's bounding surface (∂V), natively enforcing a 2D holographic projection.

The universal chaotic background possesses an irreducible Unruh temperature defined by the fundamental microscopic frequency of the vacuum equation (81). We define the macroscopic boundary of the decoherent cluster by its 2D spherical virial surface area, $A = 4\pi R_{\text{vir}}^2$. According to the Bekenstein bound [7], the number of discrete phase-space bits N that this causal boundary can support is:

$$N = \frac{Ac^3}{G\hbar}. \quad (92)$$

Following the entropic gravity formalism pioneered by Verlinde [55] and Padmanabhan [43], we evaluate the macroscopic mass equivalent of this chaotic heat through the thermodynamic equipartition of the holographic boundary. By the equipartition theorem, the total chaotic thermal energy (E_{vac})

resting on this 2D causal boundary is distributed as $\frac{1}{2}k_B T$ per bit of information:

$$E_{\text{vac}} = \frac{1}{2} N k_B T_0 = \frac{1}{2} \left(\frac{A c^3}{G \hbar} \right) k_B \left(\frac{\hbar a_0}{2\pi k_B c} \right) = \frac{a_0 c^2 A}{4\pi G}. \quad (93)$$

Converting this irreducible thermal energy to its effective dynamical mass equivalent ($M = E/c^2$) yields the total mass footprint of the chaotic vacuum bound within the cluster:

$$M_{\text{vac}} = \frac{a_0 A}{4\pi G}. \quad (94)$$

Substituting the spherical boundary area of the cluster ($A = 4\pi R_{\text{vir}}^2$), the geometric 4π factors cancel. The effective dynamical mass of the vacuum reduces to:

$$M_{\text{vac}} = \frac{a_0 R_{\text{vir}}^2}{G}. \quad (95)$$

The derivation of this effective mass from holographic equipartition aligns closely with the thermodynamic perspectives on gravity advanced by Padmanabhan [43] and the emergent dark matter formalism of Verlinde [55], where gravitational anomalies arise from the information bounds of the vacuum rather than fundamental particles. Furthermore, the emergence of a fundamental background surface density (Σ_{vac}) conceptually mirrors the universally observed constant dark matter density [21], as well as the theoretical phantom dark matter surface density limit defined in MOND [38].

Under this framework, in decoherent, pressure-supported systems, the constant surface density of the chaotic vacuum bounds the system, yielding an emergent ‘apparent dark matter’ mass that scales with the square of the virial radius ($M_{\text{vac}} \approx a_0 R_{\text{vir}}^2 / G$). For a rich cluster like Coma ($R_{\text{vir}} \approx 2$ Mpc), this estimate yields approximately $3.4 \times 10^{15} M_{\odot}$ of effective dynamical mass. Because this value appears to be of the same order as observed cluster mass discrepancies, it suggests that the missing mass problem in clusters might be interpreted not as unseen collisionless particles, but as the macroscopic holographic heat capacity of the spacetime medium.

7.2. The bullet cluster

The phase-dependent nature of Timeflow Gravity offers a possible explanation for the Bullet Cluster (1E 0657-558) [15, 33], an observation widely considered the definitive empirical proof of collisionless dark matter particles.

Observations of the Bullet Cluster show that following a high-velocity collision between two galaxy clusters, the strong gravitational lensing signal (the ‘dark matter’) is spatially separated from the bulk of the baryonic mass (the x-ray emitting intergalactic gas).

Within the present framework, this spatial separation may be interpreted not as the separation of two distinct physical substances but as the direct macroscopic signature of a shock-induced thermodynamic phase transition. During collision, the system is divided into two distinct thermodynamic states. The massive intracluster gas is collisional and experiences extreme shock heating. Its kinetic energy is randomized, resulting in a massive spike in kinematic entropy. This strongly suppresses the phase coherence of the gas ($\cos \theta \rightarrow 0$). Consequently, the structural interference term ($2k_{\text{order}} k_0 \cos \theta$) vanishes. While the gas retains its vast bare baryonic mass (k_{order}^2), the loss of phase correlation means it loses its ability to organize the surrounding vacuum background. The localized spatial gradient dissolves into the flat, unmodulated thermal background (Σ_{vac}). Because a flat background cannot lens light into a concentrated peak, the gas effectively ‘sheds’ its lensing halo.

Conversely, the constituent stars and galaxies are largely collisionless. They pass through the collision undisturbed, maintaining their low-entropy, organized kinematic trajectories. Their macroscopic phase state remains highly coherent ($\cos \theta \approx 1$). Therefore, their constructive interference term survives, continuing to phase-lock the vacuum into a sharp, localized spatial gradient ($1/r$ potential).

Because gravitational amplification in Timeflow Gravity is fundamentally an entropic response driven by the wave-mechanical interference between matter and the vacuum, the sharp spatial gradients associated with strong lensing are expected to track the low-entropy, coherent component (the galaxies) rather than the massive, decoherent component (the hot gas). The spatial separation observed in the Bullet Cluster may be interpreted as a manifestation of the spacetime medium’s phase interference. The gravitational lensing signal naturally follows the surviving order.

A quantitative test of this interpretation can, in principle, be constructed by relating the phase coherence parameter $\cos \theta$ to observable thermodynamic and kinematic properties of the cluster. In particular, the degree of coherence suppression should correlate with the local entropy and velocity dispersion

of the baryonic gas, which are directly measurable through x-ray temperature maps and spectroscopic line broadening. This suggests that the spatial offset between the lensing signal and the baryonic mass distribution could be modeled as a function of the local decoherence field, providing a predictive link between thermodynamic state and gravitational response. A detailed implementation of this program—combining hydrodynamic simulations with lensing reconstruction—lies beyond the scope of the present work and will be pursued in future studies.

8. Topological partition

To determine the global cosmological density parameters, we identify the partition between coherent and chaotic sectors with the topological geometry of the $U(1)$ phase space. We first introduce the coarse-grained statistical description of the medium, then show that its equilibrium weight is fixed by the geometric measures of the compact manifold.

The Universe contains a dynamic balance between macroscopic structures and the irreducible thermal background equation (22). We describe the global state of the spacetime medium as a coarse-grained statistical mixture of the chaotic and coherent sectors,

$$p(\phi) = (1 - f_{\text{order}}) \frac{1}{2\pi} + f_{\text{order}} p_{\epsilon}(\phi - \phi_0), \quad (96)$$

where p_{ϵ} is a sharply localized probability distribution approaching a Dirac delta function equation (17) and the dimensionless weight $f_{\text{order}} \in [0, 1]$ is the fraction of the total kinematic capacity of the medium occupied by the coherent sector.

The linear form of the mixture is the probability-distribution expression of the additive coarse-graining equation (28) already adopted at the gradient level: the field decomposes additively into an ordered mean component and a stochastic chaotic background, and the global state inherits this structure as a linear combination of the two sector distributions. Subleading correlations between sectors are carried separately by the phase-coherence cross-term of equation (62), so that the linear mixture is the unique coarse-grained description consistent with the field's existing additive decomposition.

Because energy in this framework is defined geometrically through the wavenumber, the kinematic capacity of a field excitation is fixed by its extent in the phase variable. The phase manifold is compact with total measure

$$C_{\text{total}} = \oint_{S^1} d\phi = 2\pi A_0, \quad (97)$$

where A_0 is the baseline radius of the limit cycle. This circumferential measure is the total kinematic capacity of the medium: the full extent of the manifold available to any one-dimensional excitation of the phase.

The geometry of the two sectors is fixed by the tensor structure of the field equation (62). The coherent sector enters as a rank-one form $\propto u_{\mu}u_{\nu}$, carrying a single preferred axis, while the chaotic sector enters as the isotropic form $\propto g_{\mu\nu}$, carrying no preferred direction. These two structures select the two topologically distinct classes of one-dimensional submanifold of S^1 . The isotropic mode has no axis and closes on itself, defining its full circumferential capacity equation (97). The directed mode possesses a single axis; because the rank-one form $u_{\mu}u_{\nu}$ is invariant under $u \rightarrow -u$, this axis is unoriented, and on S^1 an unoriented axis is precisely a diameter:

$$C_{\text{order}} = 2A_0. \quad (98)$$

The diameter is therefore the unique extent of the unoriented rank-one mode, attained in the fully phase-locked limit $\cos\theta \rightarrow 1$ equation (41) that defines the coherent sector. The closed loop and the straight diameter exhaust the independent one-dimensional extents of S^1 , and no third independent extent exists on the bounding manifold.

The coherent fraction is the ratio of the directed extent to the total kinematic capacity. Each is the maximal extent of its submanifold class (the open diameter and the closed loop), so the ratio is a pure number fixed by the geometry of S^1 alone:

$$f_{\text{order}} = \frac{C_{\text{order}}}{C_{\text{total}}} = \frac{2A_0}{2\pi A_0} = \frac{1}{\pi}, \quad (99)$$

independent of the limit-cycle scale A_0 . The complementary fraction $1 - f_{\text{order}} = 1 - 1/\pi$ is the isotropic background not claimed by the directed axis. Therefore, it is the portion of the circumferential capacity not occupied by the coherent projection.

Our unified field equation (62) defines the macroscopic energy density (G_{00}) as the statistical expectation value of the phase gradient:

$$G_{00}^{\text{total}} = \langle (\nabla_0 \phi_{\text{total}})^2 \rangle. \quad (100)$$

Similarly, the energy density contribution of the ordered sector is given by:

$$G_{00}^{\text{order}} = \langle (\nabla_0 \phi_{\text{order}})^2 \rangle. \quad (101)$$

Because both the ordered sector and the total medium are expectation values of the same squared phase gradient, they share a common dimensional scale k^2 . Writing the energy density of each as the product of this shared scale and a dimensionless fractional measure of its support on S^1 ,

$$G_{00}^{\text{order}} \propto k^2 C_{\text{order}}, \quad G_{00}^{\text{total}} \propto k^2 C_{\text{total}}, \quad (102)$$

the squared-gradient scale cancels exactly in the partition ratio:

$$\frac{G_{00}^{\text{order}}}{G_{00}^{\text{total}}} = \frac{k^2 C_{\text{order}}}{k^2 C_{\text{total}}} = \frac{C_{\text{order}}}{C_{\text{total}}}. \quad (103)$$

The fractional energy density reduces to the linear ratio of phase extents on S^1 . Therefore we identify the equilibrium capacity fraction with the cosmological matter density parameter:

$$\Omega_{\text{order}} \equiv f_{\text{order}} = \frac{1}{\pi} \approx 0.3183. \quad (104)$$

By the strict conservation of total capacity, the remainder of the phase space determines the baseline vacuum energy parameter:

$$\Omega_{\text{chaos}} \equiv 1 - f_{\text{order}} = 1 - \frac{1}{\pi} \approx 0.6817. \quad (105)$$

These topological derivations yield values numerically close to current cosmological measurements. The Planck 2018 [2] results measured the matter density parameter to be $\Omega_m = 0.315 \pm 0.007$.

In standard cosmology, density parameters (Ω) represent fractional 3D volumetric densities. This creates the infamous cosmological ‘coincidence problem’ [63]: it is statistically highly improbable that an evolving 3D matter density ($\propto a^{-3}$) and a static vacuum density would arbitrarily balance to the same order of magnitude only in the present epoch.

Timeflow Gravity suggests a resolution of this paradox. In this framework, the Universe is governed by a globally conserved 1D kinematic capacity. A constant global partition between coherent and incoherent states is bounded by the conserved phase space equation (11). As the 3D volumetric density of bare baryonic matter dilutes with cosmic expansion, the associated phase-locked interference contribution evolves coherently with it. Because both components arise from the same underlying phase gradient equation (62), their combined contribution preserves the global informational partition fixed by the $U(1)$ phase space, maintaining the invariant expectation value $f_{\text{order}} = 1/\pi$.

In this view, the Friedmann equation is the macroscopic 3D geometric projection of this invariant 1D topological equilibrium. This suggests that dark energy is not an independent, static cosmological fluid, but the complementary 1D phase capacity required to support the linear topological amplitude of continuous matter. Therefore, the $\approx 32/68$ split between structured energy and vacuum energy may not be an evolutionary accident of the current era, but a geometric invariant of the spacetime medium, enforced by the topology of the $U(1)$ gauge symmetry.

9. Modified Friedmann equation

In the standard cosmological model, the expansion of the Universe is described by the time evolution of a scale factor $a(t)$, governed by the Friedmann equations [24] of General Relativity. The accelerated expansion [46, 50] is conventionally framed as an intrinsic stretching of the spacetime manifold, driven in the modern era by an arbitrary cosmological constant (Λ). From the perspective of Timeflow Gravity,

it is the required macroscopic response to maintain the global conservation of information and phase space.

As established by the law of entropy equilibrium equation (24), the Universe possesses a strictly conserved total informational capacity defined by the $U(1)$ gauge symmetry of the Timeflow field. Within this conserved phase space, gravity acts fundamentally as an ordering force. As the Universe evolves from a nearly homogeneous plasma into a highly complex cosmic web of stars, galaxies, and clusters, gravity is continuously performing structural work. It concentrates matter, phase-locks the local vacuum, and drastically reduces the localized informational entropy of the spacetime medium.

Because the total informational capacity of the Universe cannot change, this continuous generation of structural negentropy must be perfectly balanced by a corresponding increase in chaotic, thermal entropy in the vacuum. In a thermodynamic system with fixed energy scale, entropy growth must therefore proceed through an increase in the number of accessible microstates, which in a field-theoretic context corresponds to an increase in the effective phase-space volume.

Therefore, in this framework, space expands in order to provide the necessary geometric capacity for the chaotic vacuum to balance the localized order generated by gravitational clustering. When the rate of structural complexity increases, space must expand proportionally to preserve the overall entropy of the spacetime medium.

This offers a possible resolution to the nature of dark energy. The large-scale cosmic acceleration is driven by the macroscopic thermodynamic heat (Q) exhausted by the continuous work (W) of gravitational structure formation. The exact same non-linear thermodynamic wave-mechanics that modulate gravity at galactic scales equation (41) also dictate the global evolution of $a(t)$.

To determine the global expansion history of the Universe, we evaluate the unified macroscopic field equation (62) within the spatially flat, homogeneous and isotropic Friedmann–Lemaître–Robertson–Walker (FLRW) [59] framework:

$$ds^2 = -c^2 dt^2 + a(t)^2 (dx^2 + dy^2 + dz^2). \quad (106)$$

where $a(t)$ is the dimensionless cosmic scale factor, normalized to $a(t_0) = 1$ at the present epoch.

Then we take our expanded field equation (62) and determine the effective cosmological energy density driving the expansion by taking the invariant scalar projection of this field equation along the comoving cosmic 4-velocity (u^μ):

$$G_{\mu\nu} u^\mu u^\nu = \left[(k_{\text{order}}^2 + 2k_{\text{order}} k_0 \cos \theta) u_\mu u_\nu - \frac{1}{4} k_0^2 g_{\mu\nu} \right] u^\mu u^\nu. \quad (107)$$

By the definition of a normalized comoving fluid in a mostly-plus metric signature $(-, +, +, +)$, the 4-velocity satisfies $u_\mu u^\mu = -1$. Consequently, the tensor contractions evaluate strictly to $u_\mu u_\nu u^\mu u^\nu = (-1)^2 = 1$, and $g_{\mu\nu} u^\mu u^\nu = -1$. For the Einstein tensor evaluated on the spatially flat FLRW metric, this invariant projection evaluates to the expansion rate:

$$G_{\mu\nu} u^\mu u^\nu = \frac{3H^2}{c^2}. \quad (108)$$

To extract the physical energy density, we project this geometric field equation along the comoving 4-velocity u^μ . Using the standard macroscopic definition of energy density,

$$\rho_{\text{total}} \equiv \frac{c^2}{8\pi G} G_{\mu\nu} u^\mu u^\nu. \quad (109)$$

and the standard 4-velocity normalization $u_\mu u^\mu = -1$, we evaluate the projection. In a strictly homogeneous and isotropic FLRW Universe, macroscopic spatial phase gradients vanish in cosmological average, $(\nabla_i \phi = 0)$, while local rank-one structures around individual matter sources persist and contribute to the cosmological mean only through their statistical averages. Therefore, the invariant wavenumber is entirely determined by the temporal evolution:

$$k_{\text{total}}^2 = -g^{\mu\nu} \langle \nabla_\mu \phi_{\text{total}} \nabla_\nu \phi_{\text{total}} \rangle = \langle (\nabla_0 \phi_{\text{total}})^2 \rangle. \quad (110)$$

Equating (108) to our field's phase variance, the total macroscopic energy density is defined strictly by the invariant wavenumber of the medium:

$$\rho_{\text{total}} = \frac{c^2}{8\pi G} \langle (\nabla_0 \phi_{\text{total}})^2 \rangle = \frac{c^2}{8\pi G} k_{\text{total}}^2. \quad (111)$$

Using the statistical decomposition of the phase field into coherent (signal) and incoherent (noise) sectors established in equation (28), the invariant wavenumber separates into three distinct thermodynamic terms equation (36). Substituting this variance sum into equation (111), we uniquely identify the physical energy density components of the Universe. The ordered component encompasses both the bare structural matter (k_{order}^2) and the phase-locked interference halo it drags out of the surrounding vacuum ($2k_{\text{order}}k_0 \cos \theta$):

$$\rho_{\text{order}} \equiv \frac{c^2}{8\pi G} (k_{\text{order}}^2 + 2k_{\text{order}}k_0 \cos \theta), \quad (112)$$

$$\rho_{\text{chaos}} \equiv \frac{c^2}{8\pi G} k_0^2. \quad (113)$$

Thus, the total energy density of the cosmos reflects the dynamic thermodynamic partition:

$$\rho_{\text{total}} = \rho_{\text{order}} + \rho_{\text{chaos}}. \quad (114)$$

The ordered component (ρ_{order}) corresponds to localized, coherent excitations of the field. These represent baryonic matter and radiation, which are strictly confined to the expanding three-dimensional spatial volume. Therefore, their bare energetic densities obey standard volumetric scaling (a^{-3} for matter, a^{-4} for radiation).

Crucially, the interference cross-term ($2k_{\text{order}}k_0 \cos \theta$) represents the ‘dark matter’ halo—the localized phase-locking of the vacuum strictly anchored to the massive coherent source. Because it is geometrically bound to the bare matter, it evolves congruently with the spatial volume of the mass distribution. We define the present-day density parameters as the sum of the bare baryonic mass (Ω_b) and this localized interference halo (Ω_{int}):

$$\Omega_m \equiv \Omega_b + \Omega_{\text{int}}, \quad (115)$$

such that the total coherent density evolves classically as:

$$\rho_{\text{order}}(a) = \rho_{c,0} \left(\frac{\Omega_m}{a^3} + \frac{\Omega_r}{a^4} \right). \quad (116)$$

To determine the chaotic component (ρ_{chaos}), we must establish how the topological partition equation (99) evolves with cosmic expansion. The framework involves two distinct partition descriptions that must be distinguished: the invariant topological capacity and its dynamically realized physical projection in expanding space. The topological partition derived in equation (99) is an invariant property of the bounding $U(1)$ manifold. By definition, the total universal phase capacity is globally conserved equation (11). However, the observable volumetric partition of energy densities evolves because the two thermodynamic sectors couple asymmetrically to the FLRW geometry.

The asymmetric coupling is fixed by the tensor structure of the field equation (62). The coherent sector enters as a rank-one form $\propto u_\mu u_\nu$, carrying a single preferred axis aligned with the local kinematic flow, while the chaotic sector enters as the isotropic form $\propto g_{\mu\nu}$, carrying no preferred direction. This tensor distinction is the field-equation expression of the same topological dichotomy that fixes the static partition: the coherent (rank-one) sector is the directed, axis-bearing projection of the $U(1)$ manifold, while the chaotic (isotropic) sector is the rotationally symmetric global background.

In the standard FLRW framework with signature $(-, +, +, +)$, the temporal metric component is invariant ($g_{00} = -c^2$), while the spatial components scale strictly with the cosmic expansion ($g_{ii} = a^2(t)$). Consequently, the proper physical length of any one-dimensional spatial projection scales precisely linearly with the scale factor $a(t)$.

Because the coherent sector enters the field equation as $\propto u_\mu u_\nu$, its physical realization is embedded along axis in the expanding spatial geometry. The directed one-dimensional structure stretches with the spatial metric, so the fraction of the invariant topological capacity dynamically realized as physical 3D structure scales linearly from its present-day ($a = 1$) boundary condition:

$$f_{\text{order}}(a) = a f_{\text{order}}(1) = \frac{a}{\pi}. \quad (117)$$

Conversely, the chaotic sector enters the field equation as $\propto g_{\mu\nu}$: a fully isotropic form, with no preferred direction onto which the asymmetric spatial metric stretching can act directly. The chaotic sector does not therefore acquire its evolution from direct metric coupling. Instead, it is depleted indirectly through the conservation of total topological capacity: as the coherent sector progressively realizes its

invariant share through spatial expansion, the available capacity remaining to the chaotic baseline diminishes correspondingly. Because the gradient decomposition equation (28) is additive and the total $U(1)$ phase capacity is a conserved invariant, the two sector fractions combine linearly, and the chaotic fraction is strictly determined by the topological remainder:

$$f_{\text{chaos}}(a) = 1 - f_{\text{order}}(a) = 1 - \frac{a}{\pi}. \quad (118)$$

This linear-in- a scaling is uniquely determined by the tensor structure of the field equation combined with the asymmetric metric of FLRW. The rank-one coherent sector couples to spatial expansion along its single preferred axis; the isotropic chaotic sector does not. Therefore, the linear form is the unique consequence of the rank-one / isotropic decomposition embedded in the expanding spatial metric.

Because the chaotic vacuum energy density is set by the squared wavenumber k_0^2 equation (113), and the chaotic fraction of the total topological capacity evolves as $f_{\text{chaos}}(a)$, the squared vacuum wavenumber inherits this scaling:

$$k_0^2(a) = k_{0,*}^2 f_{\text{chaos}}(a) = k_{0,*}^2 \left(1 - \frac{a}{\pi}\right), \quad (119)$$

where $k_{0,*}$ represents the magnitude of the baseline vacuum wavenumber at $a = 0$.

Substituting into the energy density definition yields

$$\rho_{\text{chaos}}(a) = \frac{c^2}{8\pi G} k_{0,*}^2 \left(1 - \frac{a}{\pi}\right). \quad (120)$$

Normalizing to the present-day critical density $\rho_{c,0}$, we obtain the cosmic evolution of the chaotic vacuum:

$$\frac{\rho_{\text{chaos}}(a)}{\rho_{c,0}} = 1 - \frac{a}{\pi}. \quad (121)$$

Combining the ordered structural volume and the chaotic vacuum, the total energy density of the Universe becomes a function of the scale factor a :

$$\frac{\rho_{\text{total}}(a)}{\rho_{c,0}} = \left(\frac{\Omega_m}{a^3} + \frac{\Omega_r}{a^4}\right) + \left(1 - \frac{a}{\pi}\right). \quad (122)$$

Using the standard spatial geometry relation $H^2 = \frac{8\pi G}{3} \rho_{\text{total}}$, we obtain the modified Friedmann equation for Timeflow Gravity:

$$H(a) = H_0 \sqrt{\left(\frac{\Omega_b + \Omega_{\text{int}}}{a^3} + \frac{\Omega_r}{a^4}\right) + \left(1 - \frac{a}{\pi}\right)}. \quad (123)$$

This result follows directly from the wave-mechanical structure of the Timeflow field. The ordered sector evolves according to standard volumetric dilution, while the chaotic vacuum sector evolves according to the topological constraint imposed by the finite phase-space capacity of the $U(1)$ manifold.

10. Evolving expansion

This Friedmann equation (123) suggests that dark energy is not a static cosmological constant (Λ), but it is the dynamic, depleting reservoir of the unconstrained $U(1)$ vacuum. Within this topological framework, the dark energy density must evolve and weaken over cosmic time. Remarkably, this behavior appears qualitatively consistent with recent early-stage findings of the DESI [1], while preserving standard structure formation in the early Universe.

Immediately after the Big Bang ($a \ll 10^{-4}$), the energy density of the Universe is dominated by relativistic particles (photons and neutrinos). In this limit, the radiation term Ω_r/a^4 overwhelms all other contributions in the master equation (123):

$$\frac{\Omega_r}{a^4} \gg \frac{\Omega_m}{a^3} \gg \left(1 - \frac{a}{\pi}\right). \quad (124)$$

Consequently, the expansion rate simplifies to:

$$H(a) \approx H_0 \sqrt{\frac{\Omega_r}{a^4}} = \frac{H_0 \sqrt{\Omega_r}}{a^2}. \quad (125)$$

This aligns with standard cosmology, predicting a rapid deceleration of the expansion. From the Timeflow perspective, this era represents the high-frequency regime. The energy density is so immense that the local Timeflow frequency ω is orders of magnitude higher than the vacuum baseline ω_0 . Space expands rapidly because the high energy density forces a rapid relaxation of the spacetime medium. Integrating $H = \dot{a}/a \propto a^{-2}$ yields the classic time-dependence:

$$a(t) \propto t^{1/2}. \quad (126)$$

As the Universe expands ($10^{-4} < a < 0.7$), the radiation density drops sharply (a^{-4}) due to the combined effects of volumetric dilution and redshift. The matter density, which dilutes only volumetrically (a^{-3}), eventually overtakes radiation. This crossover marks the beginning of the era of structure formation.

In this regime, the Friedmann equation (123) simplifies to:

$$H(a) \approx H_0 \sqrt{\frac{\Omega_m}{a^3}} = \frac{H_0 \sqrt{\Omega_m}}{a^{1.5}}. \quad (127)$$

This deceleration is gentler than the radiation phase. The integrated growth law becomes:

$$a(t) \propto t^{2/3}. \quad (128)$$

This recovers the standard solution for the matter dominated era.

As the Universe expands toward the present scale ($a \approx 1$), the matter density dilutes sufficiently that it no longer dominates the energy budget. The residual vacuum potential emerges as the primary driver of cosmic evolution. This marks the transition from the decelerating, structure-forming Universe to the accelerating, vacuum-dominated Universe we observe today.

In the current epoch, the radiation contribution is negligible ($\Omega_r \approx 0$). The master equation (123) is governed by the interplay between the diminishing matter density and the depleting vacuum potential:

$$H(a) \approx H_0 \sqrt{\frac{\Omega_m}{a^3} + \left(1 - \frac{a}{\pi}\right)}. \quad (129)$$

In the standard Λ CDM model, Ω_m and Ω_Λ are free parameters determined by fitting observational data. In Timeflow Gravity, however, these values are fixed by the topological partition equation (105). Recall that the matter density is geometrically constrained to $\Omega_m = 1/\pi$. Substituting this constant into the equation yields a parameter-free prediction for the expansion history:

$$H(a) = H_0 \sqrt{\frac{1}{\pi a^3} + 1 - \frac{a}{\pi}}. \quad (130)$$

The shape of the expansion curve $H(a)$ is determined by the geometry of the cycle (π), with no adjustable parameters for dark energy or matter density.

As $a \rightarrow \pi$, the vacuum energy—the engine of expansion—vanishes entirely. The unified Friedmann equation at this limit becomes:

$$H(a \rightarrow \pi) = H_0 \sqrt{\frac{\Omega_m}{\pi^3} + 0} \approx H_0 \sqrt{\frac{1/3.14}{31}} \approx 0.1 H_0. \quad (131)$$

While there is a small residual inertial expansion due to matter, the capacity of the spacetime medium to support further linear extension is exhausted.

Once the expansion halts ($\dot{a} = 0$), the dynamics are dominated by the gravitational attraction of the remaining structure and the negative pressure of the over-extended vacuum. The equation of motion reverses:

$$H(t) < 0. \quad (132)$$

The Universe enters a contraction phase. As a decreases, the vacuum potential recharges ($1 - a/\pi$ increases), and the matter density concentrates (Ω_m/a^3). This contraction leads inevitably to a ‘Big Crunch’. As $a \rightarrow 0$, the energy density spikes, eventually reaching the Planck scale.

Thus, this framework suggests the possibility of a cyclic cosmological evolution, driven eternally by the immutable geometry of the field itself.

10.1. Equation of state

A key observational test for any dark energy model is the equation of state parameter, w , defined by the ratio of pressure to density: $w = P/\rho$. In the standard Λ CDM model, the vacuum is a cosmological constant, implying a fixed value of $w = -1$. In Timeflow Gravity, the vacuum density is dynamic equation (121). We can derive the effective $w(a)$ using the continuity equation for cosmic fluids:

$$\dot{\rho}_{\text{vac}} + 3H(\rho_{\text{vac}} + P_{\text{vac}}) = 0. \quad (133)$$

The time derivative of the density is:

$$\dot{\rho}_{\text{vac}} = \frac{d\rho_{\text{vac}}}{da} \cdot \dot{a} = -\frac{\rho_{c,0}}{\pi} \cdot aH. \quad (134)$$

Substituting this into the continuity equation:

$$-\frac{\rho_{c,0}}{\pi}aH + 3H(\rho_{\text{vac}} + P_{\text{vac}}) = 0. \quad (135)$$

Dividing by $3H$:

$$\rho_{\text{vac}} + P_{\text{vac}} = \frac{\rho_{c,0}}{3\pi}a. \quad (136)$$

The equation of state parameter $w(a) = P_{\text{vac}}/\rho_{\text{vac}}$ is thus:

$$w(a) = \frac{a/(3\pi)}{1 - a/\pi} - 1 = \frac{a}{3(\pi - a)} - 1. \quad (137)$$

This result yields a prediction that dark energy is not a constant. At $a \rightarrow 0$: $w \rightarrow -1$. The early vacuum behaves like a cosmological constant. At Present ($a = 1$): Substituting $\pi \approx 3.1416$:

$$w(1) = -1 + \frac{1}{3(\pi - 1)} \approx -1 + \frac{1}{6.42} \approx -0.84. \quad (138)$$

This result identifies the vacuum as a ‘thawing’ quintessence field [11]. The acceleration of the Universe is less aggressive than a cosmological constant and is naturally braking as the cycle proceeds.

The theoretical prediction of $w(1) \approx -0.84$ is numerically very close to recent findings of evolving dark energy by the DESI Collaboration [1]. Their primary result using the DESI + cosmic microwave background (CMB) + Pantheon+ samples ($w_0 = -0.838 \pm 0.055$) aligns to the value derived here from the depletion of the vacuum, which will require detailed statistical testing in future work.

11. Falsifiability

A defining feature of General Relativity is that gravity is state-independent; the internal arrangement and kinematics of matter are largely irrelevant to its far-field flux. Timeflow Gravity directly challenges this paradigm. By identifying macroscopic gravity as an entropic response governed by the phase coherence of the underlying spacetime medium, gravitational amplification becomes fundamentally state-dependent, providing a clear observational pathway to falsify the theory.

To test Timeflow Gravity against astronomical data, we map the local phase coherence angle, θ , to measurable macroscopic observables. The phase coherence parameter $\cos\theta$ governs the macroscopic alignment of the spacetime medium with the baryonic kinematic source. In a physical system, this alignment is determined by partitioning the system’s total kinetic energy between coherent bulk motion and disordered random motion. For a baryonic distribution characterized by a coherent rotational velocity V_{rot} and an isotropic three-dimensional velocity dispersion σ_v , the kinetic energy decomposes as $K_{\text{bulk}} = \frac{1}{2}V_{\text{rot}}^2$ and $K_{\text{disp}} = \frac{3}{2}\sigma_v^2$, where the factor of three arises from the three independent spatial components of isotropic random motion. The fraction of total kinetic energy residing in coherent motion identifies directly with the phase coherence parameter:

$$\cos\theta = \frac{K_{\text{bulk}}}{K_{\text{bulk}} + K_{\text{disp}}} = \frac{V_{\text{rot}}^2}{V_{\text{rot}}^2 + 3\sigma_v^2}. \quad (139)$$

This identification follows from the wave-mechanical structure of the Timeflow field: coherent rotational motion phase-locks the local vacuum through constructive interference, while random thermal motion produces phase decoherence proportional to its kinetic energy content.

Observationally, because equation (139) can be rewritten as a pure function of the standard V/σ ratio, $\cos \theta = (V/\sigma)^2 / [(V/\sigma)^2 + 3]$, this coherence parameter can be directly evaluated using spatially resolved kinematics from standard integral field spectroscopy or radio surveys. The parameters V_{rot} and σ_v are extracted from line-of-sight velocity fields and emission-line broadening profiles using the kinematic estimators established by Cappellari [13] and Lelli *et al* [32]. This geometric partitioning of dynamical regimes parallels the interpolating angle $\cos \gamma_s$ utilized in the HMG [41], which partitions between the system's gravitational binding and its cosmic expansion context.

To evaluate this framework against the RAR [34], we isolate the observable radial component of the wave-mechanical equilibrium function equation (87). We achieve this by exploiting the geometric structure of μ and the symmetry properties of the isotropic vacuum background.

Setting $a = g_{\text{obs}}$ as the kinematic acceleration of the system, the equilibrium function takes the form:

$$\mu(g_{\text{obs}}, a_0, \theta) = \frac{g_{\text{obs}}^2 + g_{\text{obs}} a_0 \cos \theta}{g_{\text{obs}}^2 + a_0^2 + 2g_{\text{obs}} a_0 \cos \theta}. \quad (140)$$

The denominator admits a natural geometric decomposition. Using the identity $\cos^2 \theta + \sin^2 \theta = 1$, we rewrite:

$$g_{\text{obs}}^2 + a_0^2 + 2g_{\text{obs}} a_0 \cos \theta = (g_{\text{obs}} + a_0 \cos \theta)^2 + a_0^2 \sin^2 \theta. \quad (141)$$

This decomposition reveals that the total acceleration magnitude decomposes into a radial component, $(g_{\text{obs}} + a_0 \cos \theta)$, aligned with the coherent matter gradient, and a tangential component, $a_0 \sin \theta$, representing the isotropic vacuum background orthogonal to the radial direction. The equilibrium function correspondingly becomes:

$$\mu = \frac{g_{\text{obs}} (g_{\text{obs}} + a_0 \cos \theta)}{(g_{\text{obs}} + a_0 \cos \theta)^2 + a_0^2 \sin^2 \theta}. \quad (142)$$

The tangential component represents the uniform, isotropic vacuum baseline. By Newton's Shell Theorem, a spherically symmetric isotropic background exerts no net radial force on a test particle interior to it. Therefore, while the tangential component contributes to the local thermodynamic state of the medium, it produces no observable radial acceleration. Eliminating this isotropic contribution from the denominator, the effective radial equilibrium function reduces to:

$$\mu_{\text{radial}} = \frac{g_{\text{obs}} (g_{\text{obs}} + a_0 \cos \theta)}{(g_{\text{obs}} + a_0 \cos \theta)^2} = \frac{g_{\text{obs}}}{g_{\text{obs}} + a_0 \cos \theta}. \quad (143)$$

The macroscopic relationship between the bare baryonic source and the total observed acceleration is $g_{\text{bar}} = g_{\text{obs}} \cdot \mu$, derived from the modified field equation in the weak-field limit equation (79). Substituting the radial equilibrium function:

$$g_{\text{bar}} = g_{\text{obs}} \cdot \mu_{\text{radial}} = \frac{g_{\text{obs}}^2}{g_{\text{obs}} + a_0 \cos \theta}. \quad (144)$$

Multiplying through by the denominator and rearranging yields a quadratic relationship between the observed and bare baryonic accelerations, parameterized by the local phase coherence $\cos \theta$:

$$g_{\text{obs}}^2 - g_{\text{bar}} g_{\text{obs}} - g_{\text{bar}} a_0 \cos \theta = 0. \quad (145)$$

Solving for the positive root provides the radial field equation of Timeflow Gravity:

$$g_{\text{obs}} = \frac{g_{\text{bar}} + \sqrt{g_{\text{bar}}^2 + 4g_{\text{bar}} a_0 \cos \theta}}{2}. \quad (146)$$

This single equation establishes absolute mathematical boundaries based on the thermodynamic state of the source. In rotationally supported spiral galaxies ($V_{\text{rot}} \gg \sigma_v$), phase coherence approaches unity ($\cos \theta \approx 1$). The equation maximizes thermodynamic efficiency, resolving exactly to the standard interpolating function and reproducing the universal RAR. Conversely, in diffuse, high-entropy systems such as ultra-diffuse galaxies where $\sigma_v \gg V_{\text{rot}}$, phase coherence breaks down ($\cos \theta \rightarrow 0$). The macroscopic dark matter halo naturally evaporates ($g_{\text{obs}} = g_{\text{bar}}$), resolving the missing mass discrepancy in galaxies like NGC 1052-DF2 [20] without requiring fine-tuned dark matter.

This state-dependence leads to a specific, testable prediction: the intrinsic scatter of the RAR is not random observational noise, but correlates strictly with the kinematic disorder of the source. We define the entropic residual δ as the deviation from the perfectly coherent universal response (g_{RAR}):

$$\delta = \log(g_{\text{obs}}) - \log(g_{\text{RAR}}). \quad (147)$$

Therefore, Timeflow Gravity presents a definitive falsification criterion: the entropic residuals (δ) must exhibit a strict anti-correlation with the local thermodynamic state. If the far-field gravitational acceleration is observed to be strictly a function of enclosed mass regardless of the system's internal velocity dispersion, the hypothesis is falsified. If, however, the residuals of the RAR correlate exactly with the macroscopic phase angle ($\cos\theta$), the state-dependent nature of gravity is confirmed.

12. Future directions and discussion

The aim of this work is not to construct a fundamental theory of quantum gravity, but to demonstrate that the thermodynamic and informational principles underlying the Timeflow Gravity framework are sufficient to account for a wide range of gravitational phenomena across scales. In particular, we show that local galactic-scale anomalies and the accelerated cosmic expansion can be unified within a single macroscopic description, without the introduction of arbitrary free parameters.

By treating gravity not as a fundamental interaction, but as an emergent entropic response of the $U(1)$ spacetime medium, this framework provides a consistent interpretation in which the effects attributed to dark matter and dark energy arise as complementary manifestations of the vacuum's thermodynamic partition.

The immediate next step for this framework is to explicitly test its modified Friedmann equation (123) against the standard cosmological probes. Specifically, the dynamic depletion of the vacuum must be examined against the latest luminosity distance-redshift relations of Type Ia Supernovae [10] and the early-stage evolving dark energy models suggested by recent DESI data [17].

Furthermore, the framework must be tested to the physics of the early Universe. The invariant topological matter density equation (104) must be reconciled with the acoustic peaks of the CMB power spectrum [2]. Additionally, exploring how the bounded $U(1)$ phase space behaves at the extreme $a \rightarrow 0$ limit may provide a natural thermodynamic mechanism for an emergent alternative to the epoch of cosmic inflation [25].

Observationally, the falsifiability criteria laid out in this work provide a strict, near-term pathway to test the framework. By explicitly linking the magnitude of gravitational amplification to the phase coherence angle (θ) of the source matter, the theory predicts a specific anti-correlation between the RAR [34] residuals and local kinematic entropy. Executing this test against high-resolution integral field unit surveys [4] will definitively falsify the state-dependent nature of gravity.

For now, the theory in its current state is classical and macroscopic in nature. The next stage of theoretical development must focus on building a rigorous Lagrangian density for the Timeflow field, from which the field equations can be formally derived via the action principle, and which allows for a consistent covariant coupling to matter and radiation. Such a formulation holds the potential to quantize the theory, opening the door to a complete framework of quantum gravity grounded in these foundational thermodynamic principles.

13. Conclusion

General Relativity and the Λ CDM model have achieved remarkable success in describing gravitational phenomena across a wide range of scales. However, this success relies on the introduction of dark matter and dark energy as independent components whose physical origin remains unknown. In this sense, the standard model provides an accurate description of what is observed, but does not yet offer a fundamental explanation of why these phenomena arise.

Timeflow Gravity offers a different perspective. In this work, we develop a unified framework in which gravitational dynamics emerge from the thermodynamic and wave-mechanical structure of a conserved $U(1)$ phase field equation (1). The central principle of the theory is that spacetime possesses a finite, invariant phase space equation (11), whose total informational capacity is strictly conserved and dynamically partitioned between coherent (ordered) and incoherent (thermal) components.

Starting from the time-energy equivalence principle and the law of entropy equilibrium, we constructed a macroscopic description of spacetime in which gravitational effects arise from this partition. Within this framework, we derive a modified Einstein field equation (62) in which the effective gravitational response is regulated by a wave-mechanical efficiency factor μ equation (41), encoding the local thermodynamic state of the spacetime medium. This formulation yields several key results.

In the weak-field limit, the theory naturally reproduces MOND for coherent, low-entropy systems, providing an origin for the characteristic acceleration scale a_0 . At the same time, the framework suggests a resolution of the long-standing cluster-scale discrepancy by showing that gravitational amplification explicitly depends on phase coherence. The theory also provides a natural interpretation of the Bullet Cluster. The observed separation between baryonic matter and gravitational lensing peaks is explained as a shock-induced thermodynamic phase transition: collisional gas becomes decoherent and loses its ability to organize the surrounding vacuum, while collisionless components remain phase-coherent and continue to generate the gravitational signal.

At cosmological scales, the conserved phase space of the $U(1)$ field imposes a strict global partition between order and chaos. This leads to a parameter-free prediction for the present-day matter density equation (104), in agreement with observational constraints. The complementary vacuum component emerges as the residual phase capacity, yielding a dynamical dark energy density equation (105).

From this topological constraint, we derived a modified Friedmann equation (123) that reproduces the standard radiation and matter-dominated eras while predicting a late-time accelerating phase driven by the thermodynamic response of the vacuum. Crucially, the theory predicts an evolving dark energy equation of state equation (137), with $w \approx -0.84$ at the present epoch, consistent with recent DESI [1] observations.

Taken together, these results suggest that the phenomena currently attributed to dark matter and dark energy may be understood as emergent consequences of a single underlying principle: the conservation and partition of the phase space of spacetime. In this view, gravity is not a fundamental interaction, but a macroscopic manifestation of the statistical mechanics of a bounded phase field.

A particularly interesting feature of this approach is the environment-dependent drive of energy to preserve ordered states. This principle may extend far beyond gravitation alone, potentially offering a universal mechanism for the emergence of complex structures. From particle-scale stability to large-scale structure formation, this thermodynamic principle may provide a unifying framework for understanding the emergence of ordered systems across physical scales.

This unification opens a path toward a deeper theory, where matter, gravity, space and time are not isolated phenomena but representations of one continuous thermodynamic process. This idea therefore holds the potential not only to resolve the puzzles of dark matter and dark energy, but also to illuminate the fundamental principles that shape the evolution of the Universe.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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